DIGITAL TWIN CHALLENGE 2026 • HAPPIEST HEALTH

GlucoTwin

A digital twin prototype for Type 2 Diabetes care support

Dual-stream EHR + simulated wearable data • Glycaemic event risk estimates • Clinician review workflow

Team: [ADD OFFICIAL TEAM NAME] **Institution:** [ADD COLLEGE / INCUBATOR]

Team leader: Ayush Gupta **Members:** [ADD ALL MEMBERS]

SYNTHETIC DATA ONLY — NOT FOR CLINICAL USE

Why a digital twin?

Disconnected context Clinical history, labs, medications and wearable trends may be reviewed separately	**Fast-changing glucose** Glycaemic excursions can develop over hours between appointments	**Review workload** Care teams need a consolidated view of trends, estimates and investigations to review

Project question: can a patient-specific digital-twin workflow combine these streams and surface useful review prompts?

What GlucoTwin is designed to do

1. INGEST	2. FUSE	3. ESTIMATE	4. REVIEW
Synthetic EHR context and simulated CGM / wearable time series	Engineer time-window features and combine them with clinical context	LightGBM estimates event risk over a nominal 120-minute horizon	Display outputs, investigation prompts and audit events for clinician review

The prototype supports review; it does not automate clinical decisions.

System design

STREAM A — SYNTHETIC EHR Demographics, diagnoses, HbA1c/eGFR, medications and comorbidities	**STREAM B — SIMULATED WEARABLES** CGM, HRV, sleep, steps and recent trends
VALIDATION + FEATURE ENGINEERING Quality checks, windows, missingness and derived features	**PERSISTENT TWIN STATE** Recent trends, risk history, last-test dates and clinician actions
EARLY FUSION → LIGHTGBM Hyperglycaemia and hypoglycaemia estimates; nominal 2-hour horizon	**VERSIONED RULES + EXPLANATIONS** Review prompts, provenance and contribution summaries where configured
FASTAPI SERVICE Typed endpoints, prediction/context payloads and audit events	**REACT CLINICIAN DASHBOARD** Review and accept/defer/override audit trail

Clinicians retain decision authority. Outputs do not order tests, change medication or prescribe treatment.

Technical stack

Area	Technology	Purpose
Data	Python, NumPy, SciPy, Pandas, Parquet	Synthetic EHR / wearable simulation
ML	LightGBM, scikit-learn	Tabular prediction and evaluation
Explainability	SHAP / contribution layer where configured	Interpretability support
API	FastAPI, Pydantic, Uvicorn	Typed service endpoints
Frontend	React, TypeScript, Vite, Tailwind, Recharts	Clinician-facing dashboard
State & QA	PostgreSQL, SQLite, pytest, Docker Compose	State, tests and packaging

FHIR R4 mapping is a design direction; real integration requires implementation and validation.

Current generated results

Target	AUROC	AUPRC	Brier	Samples
Hyperglycaemia	0.9192	0.8277	0.0652	54,946
Hypoglycaemia	0.5000	0.0192	0.0192	114,440

Headline metrics are from metrics.json. Full-fusion ablation results differ (hyper AUROC 0.9789; hypo AUROC 0.8264). Reconcile the evaluator before performance claims.

Synthetic-data results are not clinical validation or proof of real-world generalisation.

Feature-set comparison

Target	Feature set	AUROC	AUPRC
Hyper	EHR only	0.8538	0.6414
Hyper	CGM only	0.9520	0.8802
Hyper	CGM + wearable	0.9786	0.9408
Hyper	Full fusion	0.9789	0.9414
Hypo	EHR only	0.7260	0.0342
Hypo	CGM only	0.6366	0.0371
Hypo	CGM + wearable	0.7154	0.0446
Hypo	Full fusion	0.8264	0.0616

Safety and privacy

Synthetic only Use generated records and simulated signals. Do not commit real patient data.	**Human in the loop** Investigation prompts are suggestions, not orders. No automated diagnosis, prescribing or therapy changes.	**Validation required** Clinician review, independent validation, leakage checks, calibration, privacy/security and regulatory assessment are prerequisites.

Current limitations

- Synthetic-only evaluation does not establish real-world performance.
- Headline and ablation metrics disagree and must be reconciled.
- Lead-time interpretation requires verification.
- Investigation rules are pending clinician review.

Summary & submission

GlucoTwin explores how longitudinal clinical context and dynamic sensor data can be combined into a reviewable patient-specific digital-twin workflow.

Team: [ADD OFFICIAL TEAM NAME] **Institution:** [ADD COLLEGE / INCUBATOR]

Team leader: Ayush Gupta **Members:** [ADD ALL MEMBERS]

Repository: https://github.com/ayush2459/medai

Thank you

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